Engine Model(s): PT6A

77-20-01

T5 INDICATING SYSTEM - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00 STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, 0.025 inch diameter, and will not be specified in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC05-066	Tape
PWC05-101	Cloth, Abrasive
PWC05-110	Lockwire
PWC05-164	Welding Material, Alumel
PWC05-353	Welding Material, Chromel
PWC06-023	Compound, Anti-seize
PWC08-009	Cement, Filler and Binder

3. Special Tools

The special tools listed below are used in the following procedures.

Tool No.	Name
PWC30114-08	Wrench
PWC30425	Pliers, Crimping

4. Fixtures, Equipment and Supplier Tools

The fixtures, equipment and supplier tools listed below are used in the following procedures.

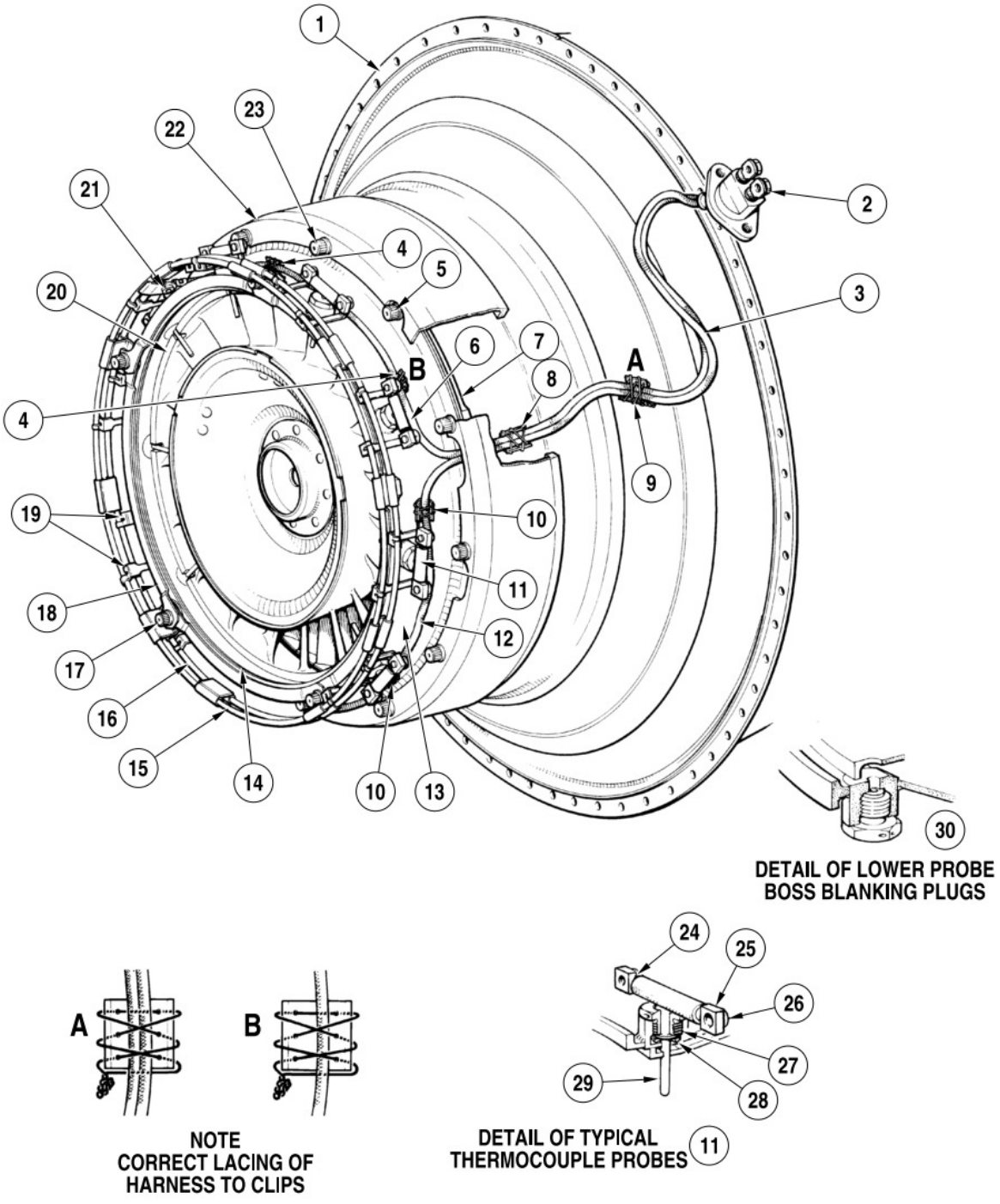
Name	Remarks
Test Set	Barfield, TT1000A or TT1200A

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5. Removal/Installation

A. Removal of Bus-bar and Probe Assembly (Ref. Fig. 201)

Figure 201 Removal/Installation of Bus-Bar Assembly and Thermocouple Probes



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1. Exhaust Duct
2. T5 Terminal Block
3. T5 Thermocouple Harness
4. Clip
5. Bolt (Half Ring Retaining)
6. Long Lead
7. Half Rings
8. Clip
9. Clip
10. Clip
11. Thermocouple Probe
12. Short Lead
13. Stator Housing
14. Interstage Sealing Ring
15. Alumel Bus-bar
16. Chromel Bus-bar
17. Screws
18. Retaining Plates
19. Bus-bar Straps
20. Power Turbine Vane Ring
21. Socket Head Screw
22. Containment Ring
23. Bolt
24. Alumel Terminal
25. Chromel Terminal
26. Capscrew
27. Coupling Nut
28. Copper Washer
29. Thermocouple Probe
30. Plugs (2 of)

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(1) Remove power section (Ref. 72-00-00).

CAUTION: USE CARE WHEN THE TERMINAL SCREWS ARE REMOVED FROM PROBES. INSULATION MATERIAL IS EXTREMELY FRAGILE.

(2) Disconnect wiring harness (3):

(a) Remove screws (21) which connect leads of harness to respective terminal (15 and 16) on bus-bar assembly.

(b) Detach long lead (alumel) (6) of harness from clips (4), and short lead (chromel) (12) from clips (10) on stator housing (13).

(c) Fold leads forward and tape to containment ring (22) to prevent damage to leads.

(3) Remove capscrews (26, Pre-SB1503) or bolts (26, Post-SB1503) which connect the respective bus-bar straps (19) to terminals of each probe (11).

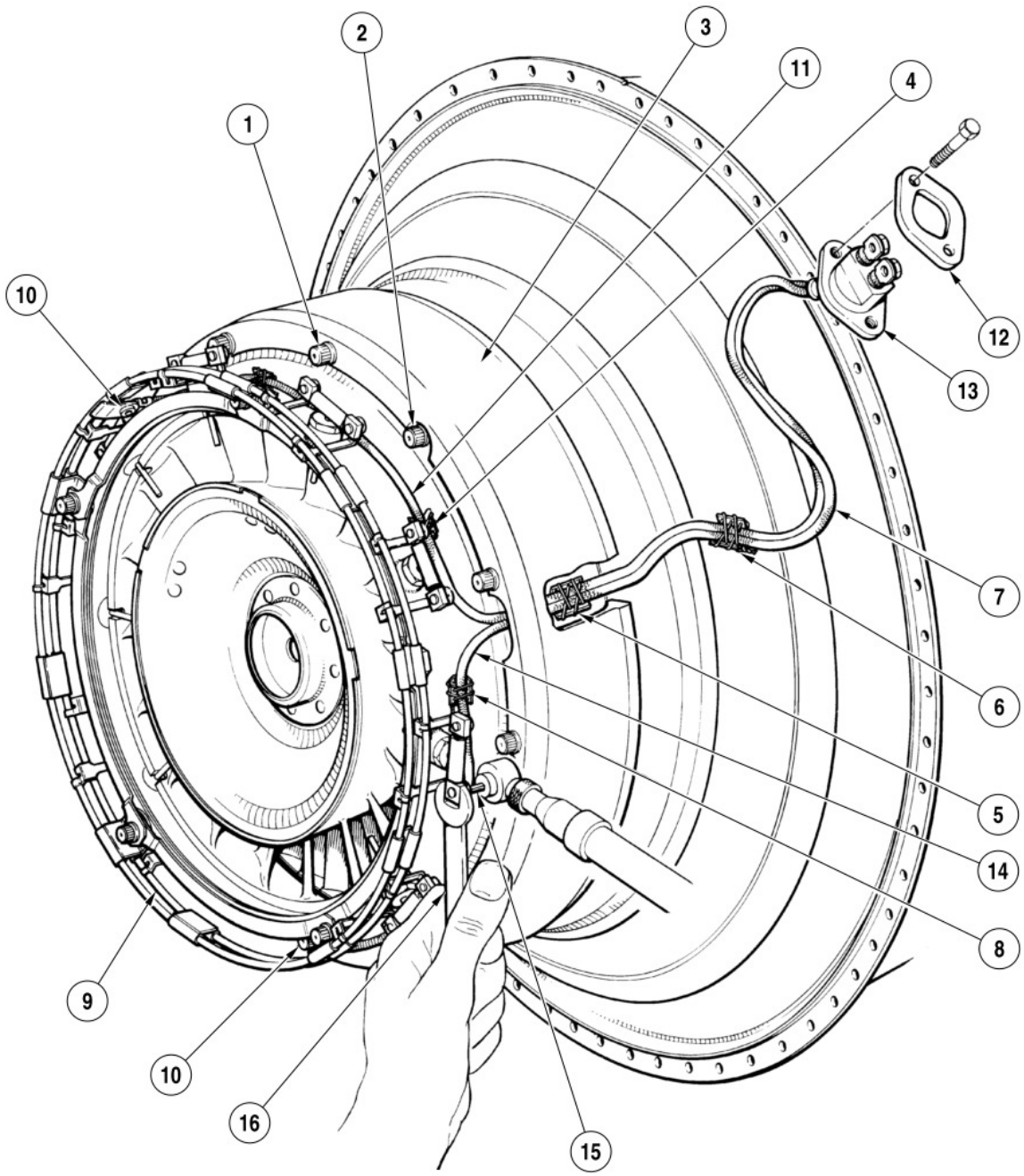
(4) Remove five screws (17), the retaining plates (18) and the bus-bar assembly from the stator housing (13).

(5) Use wrench (PWC30114-08) to loosen coupling nut (27) of each probe (11). Remove probes. Discard washers (28).

B. Removal of Wiring Harness (Ref. Fig. 202)

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Figure 202 Removal/Installation of T5 Wiring Harness



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1. Bolt
2. Bolt
3. Containment Ring
4. Clip (Stator Housing - Alumel Lead)
5. Clip (Exhaust Duct)
6. Clip (Exhaust Duct)
7. T5 Wiring Harness
8. Clip (Stator Housing -Chromel Lead)
9. Bus-bar Assembly
10. Capscrews
11. Long Lead (Alumel)
12. Gasket
13. Terminal Block
14. Short Lead (Chromel)
15. Allen Wrench (Pre-SB1503) or Socket (Post-SB1503)
16. Open-ended Wrench

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(1) Disconnect airframe leads and T1 trim harness leads from terminal block (13) at 2 o'clock position on gas generator case.

(2) Remove power section (Ref. 72-00-00). When the terminal block of wiring harness is detached, discard gasket (12).

(3) Remove capscrews (10) and disconnect leads (11 and 14) of wiring harness from respective terminal on bus-bar assembly (9).

(4) Detach long lead (11) of harness from clips (4), and short lead (14) from clips (8) on stator housing.

(5) Remove eight bolts (1) which secure containment ring (3) to Flange D of exhaust duct.

NOTE: Do not remove bolts (2) at 1 and 7 o'clock positions on flange of exhaust duct. These bolts secure the stator housing to the duct.

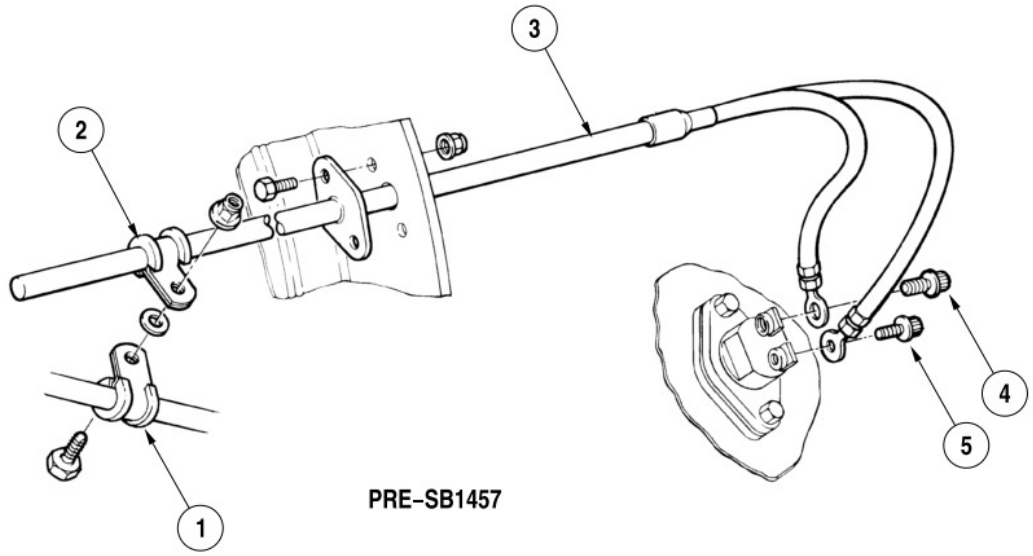
(6) Move the containment ring (3) rearward (use care to prevent damage to thermocouple probes) sufficiently to clear clip (5) on exhaust duct.

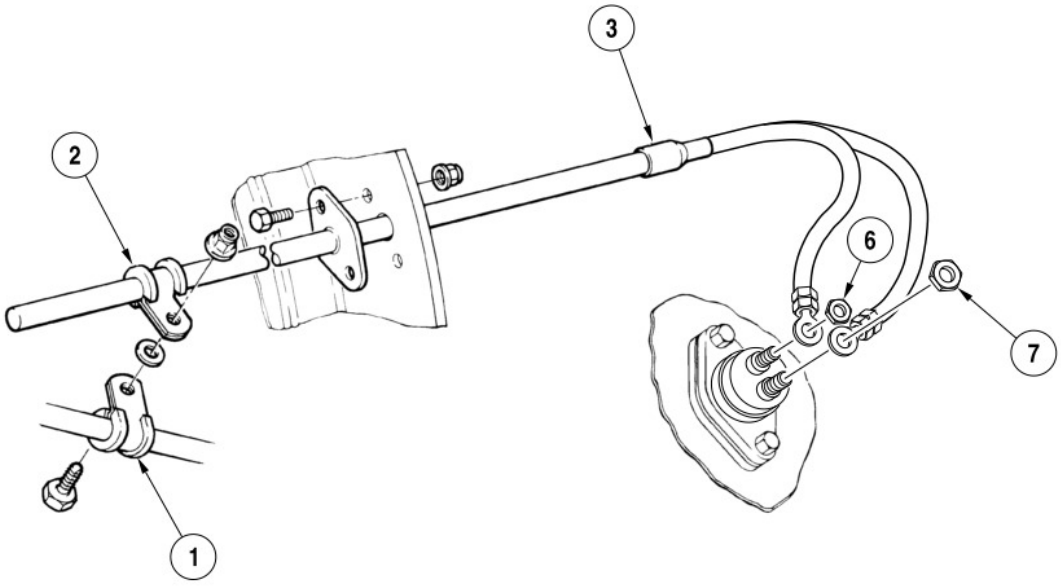
(7) Detach harness (7) from clips (5 and 6) on exhaust duct. Carefully withdraw harness clear of containment ring (3).

C. Removal of Trim Thermocouple (Ref. Fig. 203)

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Figure 203 Removal/Installation of Trim Thermocouple





Post-SB1457

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1. Clamp
2. Clamp
3. Trim Thermocouple
4. Bolt (Alumel Terminal)
5. Bolt (Chromel Terminal)
6 Self-locking Nut (Post-SB1457)
7 Self-locking Nut (Post-SB1457)

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(1) For Pre-SB1457 Engines : Remove the bolts (4 and 5) attaching the leads of the trim thermocouple (3) and the airframe leads to the posts on the terminal block mounted on the gas generator case.

(2) For Post-SB1457 Engines : Remove the nuts (6 and 7) attaching the leads of the trim thermocouple (3) and the airframe leads to the studs on the terminal block mounted on the gas generator case.

(3) Remove the self-locking nut, washer and bolt securing the clamp (2) to the clamp (1) on the wire rope casing. Remove the clamp (2).

(4) Remove the two self-locking nuts and bolts securing the flange of the trim thermocouple to the center fireseal mount ring.

(5) Remove the trim thermocouple by withdrawing rearward through the center fireseal mount ring. Exercise care not to chafe the leads or damage the terminal lugs.

D. Installation of Bus-bar and Probe Assembly (Ref. Fig. 201)

(1) Install new copper washer (28) on each termocouple probe (11).

CAUTION: APPLY ANTISEIZE COMPOUND TO THERMOCOUPLE PROBE THREADS ONLY.

(2) Apply anti-seize compound (PWC06-023) to the T5 thermocouple probe threads only with a small paintbrush.

NOTE: Do not use engine oil for a lubricant in this area.

(3) Install probes (11) in bosses on stator housing (13). Use wrench to tighten coupling nuts (27) and torque 40 to 50 lb.in. Make sure terminal bar of each probe is parallet to rear flange of stator housing and lockwire coupling nuts

NOTE: Before the bus-bar is installed, clean contact surfaces of straps and terminals, and mating surfaces on probes and wiring harness.

(4) Position five retaining plates (18) on top of the bus-bar mounting brackets so that the ends overlap the bracket and each other.

(5) Locate the bus-bar assembly on the front flange of the stator housing (13) and align the mounting holes. Secure the bus-bar assembly and the retaining plates (18) to the stator housing with five screws (17). Tighten the screws and torque to 22 to 24 lb.in. Secure with lockwire.

NOTE: Use care when the straps (19) are positioned against the respective terminals (24 and 25) of the probes.

CAUTION: USE CARE WHEN THE TERMINAL SCREWS OF PROBES ARE INSTALLED. INSULATION MATERIAL IS EXTREMELY FRAGILE.

(6) Secure straps (19) of bus-bar assembly to terminals (24 and 25) of probes with respective capscrews (26, Pre-SB1503). or bolts (26, Post-SB1503).

(a) For Pre-SB1503 Engines: Use a 9/64 inch Allen wrench to capscrews at AL terminals and a 7/64 inch Allen wrench at CR terminals. Torque AL and CR terminal capscrews 9 to 12

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lb.in. and lockwire.

(b) For Post-SB1503 Engines: Torque AL terminal bolts 8 to 12 lb.in. and lockwire. Torque CR terminal bolts 8 to 10 lb.in. and lockwire.

(7) Route long lead (alumel) (6) of T5 wiring harness along top of stator housing and connect to bus-bar terminal at approximately the 10 o'clock position with screw (21, Pre-SB1503) or bolt (21, Post-SB1503). Route short lead (chromel) (12) along lower side of housing and connect to bus-bar terminal at approximately the 5 o'clock position with screw (21, Pre-SB1503) or bolt (21, Post-SB1503).

(a) For Pre-SB1503 Engines: Torque AL and CR terminal screws 9 to 12 lb.in. and lockwire .

(b) For Post-SB1503 Engines: Torque AL terminal bolt 8 to 12 lb.in. and lockwire. Torque CR terminal bolt 8 to 10 lb.in. and lockwire .

(8) Lockwire leads of harness to clips (4 and 10).

(9) Do a functional test (Ref. 77-20-00).

(10) Install the power section (Ref. 72-00-00).

NOTE: A performance check will be done at engine test (Ref. 71-00-00).

E. Installation of Wiring Harness (Ref. Fig. 202)

NOTE: Clean terminal lug contact surfaces before installation.

(1) Carefully pass leads of harness (7) under containment ring (3) and pull rearward. Use care to not chafe leads.

(2) Secure containment ring to Flange D of exhaust duct with one bolt at 12 o'clock position. Fingertighten bolt.

(3) Route long lead (11) of harness along top of stator housing and connect to alumel terminal of bus-bar assembly (9) with capscrews (10, Pre-SB1503) or bolts (10, Post-SB1503). Route short lead (14) of harness along lower side of housing and connect to chromel terminal of bus-bar with capscrew (10, Pre-SB1503) or bolt (10, Post-SB1503).

(a) For Pre-SB1503 Engines: Use a 9/64 inch Allen wrench for capscrew at AL terminal and a 7/64 inch Allen wrench at CR terminal. Torque AL and CR terminal capscrews 9 to 12 lb.in. and lockwire.

(b) For Post-SB1503 Engines: Torque AL terminal bolt 8 to 12 lb.in. and lockwire. Torque CR terminal bolt 9 to 10 lb.in. and lockwire.

(4) Secure leads (11 and 14) to respective clips (4 and 8) on stator housing with lockwire.

CAUTION: USE CARE TO NOT DAMAGE THERMOCOUPLE PROBES OR CHAFE HARNESS LEADS.

(5) Remove slave bolt at 12 o'clock position and move containment ring rearward sufficiently to clear clip (5) on exhaust duct.

(6) Secure both leads of harness to clips (5 and 6) on exhaust duct with lockwire.

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(7) Secure containment ring (3) to flange of exhaust duct with bolts (1). Tighten bolts and torque 32 to 36 lb.in. Lockwire bolts (1 and 2).

(8) Install power section (Ref. 72-00-00). Use new gasket (12) when terminal block of wiring harness is attached to gas generator case.

(9) Do a functional test (Ref. 77-20-00).

(10) Connect T1 trim harness leads ad airframe leads to terminal block.

(11) Performance check T5 sensing system at engine test (Ref. 71-00-00).

F. Installation of Trim Thermocouple (Ref. Fig. 203)

NOTE: 1. Prior to installation, make sure that the terminal lug contact surfaces are cleaned (Ref. Para. 8.).

NOTE: 2. Should the T5 trim harness require replacement in the field, it must be replaced with one having identical resistance. Careful note should be made of the part number of the unit removed, and the part number of the unit to be installed (Ref. IPC).

(1) Install the trim thermocouple (3) by passing the leads through the center fireseal mount ring from the rear. Secure the flange of the trim thermocouple to the fireseal mount ring with two bolts (heads on the rear face of the fireseal mount ring) and self-locking nuts. Tighten the nuts 36 to 40 lb.in.

(2) Install the clamp (2) on the probe of the thermocouple and fasten to the clamp (1) on the wire rope casing with a bolt, washer (between the clamps) and self-locking nut. Tighten the nut 36 to 40 lb.in.

NOTE: Align the clamps without exerting bending pressure on the probe.

(3) For Pre-SB1457 Engines : Connect the leads of the trim thermocouple and the airframe leads to the respective AL and CR posts on the terminal block with the bolts (4 and 5). Tighten bolts and torque bolt (4) 10 to 15 lb.in., and bolt (5) 8 to 10 lb.in.

(4) For Post-SB1457 Engines: Connect the leads of the trim thermocouple and the airframe leads to the respective AL and CR studs on the terminal block with nuts (6) and (7). Tighten nuts. Torque the (large nut) 10 to 15 lb.in. Torque the small nut (6) 8 to 12 lb.in.

6. Adjustment/Test

A. Bus-bar and Probe Assembly

NOTE: This test is done with the thermocouple at room temperature. The probe being tested must be removed from the bus-bar.

(1) Using the test set, measure the insulation resistance across either the alumel or chromel terminal of the probe and ground (vane housing). The resistance measured must be not less than 50,000 ohms.

(2) Measure and record the continuity resistance of the probe loop by connecting the leads of the test set to the thermocouple terminals.

(3) Switch the test set leads to the opposite terminals on the thermocouple probe, measure and record the continuity and loop resistance.

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NOTE: Readings must be taken in quick succession to keep the temperature variation to a minimum.

(4) Add the two recorded values and divide by two to find the average loop resistance.

(5) Acceptable resistance is 0.24 to 0.46 ohm.

CAUTION: DO NOT ALLOW THE PROBE TEMPERATURE TO EXCEED TO 538°C (1000°F) DURING THE FOLLOWING TEST.

(6) When performing the heat response check of the probe, confirm that the output signal is of the same polarity as that of other the probes. Connect the test set to the AL and CR terminals, apply heat to each probe in turn and check for a positive indication of temperature change. A heat source, such as a 100 watt soldering iron, capable of heating the individual probes to 94°C (200°F) can be used.

NOTE: A heat response check of probes installed and connected to the bus-bar assembly, may be performed provided the previously heated probe is allowed to cool before applying heat to the next probe.

B. Wiring Harness

NOTE: Make sure the harness is dry and free of moisture before proceeding with the test.

(1) Use the test set and with busbar and probe assembly disconnected from the wiring harness, measure the insulation resistance between the alumel and chromel lugs and between individual lugs and wiring harness shield. Resistance measured should not be less than 10 MΩ.

(2) Check for continuity of the alumel and chromel lead of the wiring harness. Continuity must prevail for both leads. If the leads fail the test, or are found shorted, proceed as follows:

(a) Remove any moisture and heat the T5 terminal block for five minutes using a heat gun having a maximum heat range of 538°C (1000°F).

(b) Let the terminal block cool to room temperature.

(c) Repeat the continuity test.

(3) If test in steps (1) or (2) is not satisfactory, replace the wiring harness.

C. Trim Thermocouple

(1) Using the test set, with the thermocouple disconnected, measure the insulation resistance between either the alumel or chromel lug and the thermocouple outer casing. The resistance measured must be not less than 50 000 ohms.

(2) Measure the resistance of the probe at room temperature; the resistance must be within plus or minus five percent of the value for the relevant class (Ref. Table 201).

Table 201 Trim Thermocouple Conversion Table

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P/N 3013604 and class	Resistance Value (Ohms at 70 °F)	P/N 3031417 and class
	3.75 to 3.95	10
	3.95 to 4.20	11
	4.20 to 4.45	12
	4.45 to 4.75	13
	4.75 to 5.10	14
	5.10 to 5.50	15
	5.50 to 5.90	16
	5.90 to 6.40	17
	6.40 to 6.90	18
	6.90 to 7.60	19
	8	20
1	9	25
2	10.5	30
3	12	35
5	14	40
7	16	45
10	19	50
20	24	55
30	30	60
40	43	65
50	72	70
60	110	75

7. Inspection/Check

A. Inspection

(1) Inspect the installed bus-bar probe assembly for security and signs of damage. Damaged probes will necessitate individual replacement. Loose terminal posts and missing insulation may be repaired (Ref. Para. 9.).

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(2) T5 wiring harness inspection:

(a) Examine the harness for broken or damaged braiding and/or insulation. Stainless braiding that is lifted or broken may be cut away, provided insulation remains intact.

(b) Examine the terminal block for security to the harness leads, loose terminals and missing insulation. Replace missing insulation and secure loose terminals (Ref. Repair).

(c) Examine terminal lugs for condition. Replace loose or damaged lugs (Ref. Repair).

(3) Trim thermocouple inspection:

(a) Inspect the trim thermocouple leads for damage to the outer covering and loose or damaged terminal lugs. Check the probe outer casing for damage and corrosion, and the flange for security.

(4) Bus-bar inspection:

(a) Bus-bar insulation resistance must not be less than 50 000 ohms.

(b) Check bus-bar for breaks, kinks, damage or loose bracket.

(c) Return defective bus-bar for possible repair to the following P&WC facility:

P&WC Accessories Services
333 Rue D'Auvergne (Area 2K)
Longueuil, Quebec
Canada
J4H 3Y3

8. Cleaning/Painting

A. Cleaning

(1) Clean the probes that exhibit carbon deposits with a fine brass wire brush.

(2) Clean the contact surfaces of the AL and CR terminals and all harness lugs with No. 400 grit abrasive cloth (PWC05-101) .

(3) Clean the terminal block and contact surfaces of the harness lugs with No. 400 grit abrasive cloth (PWC05-101).

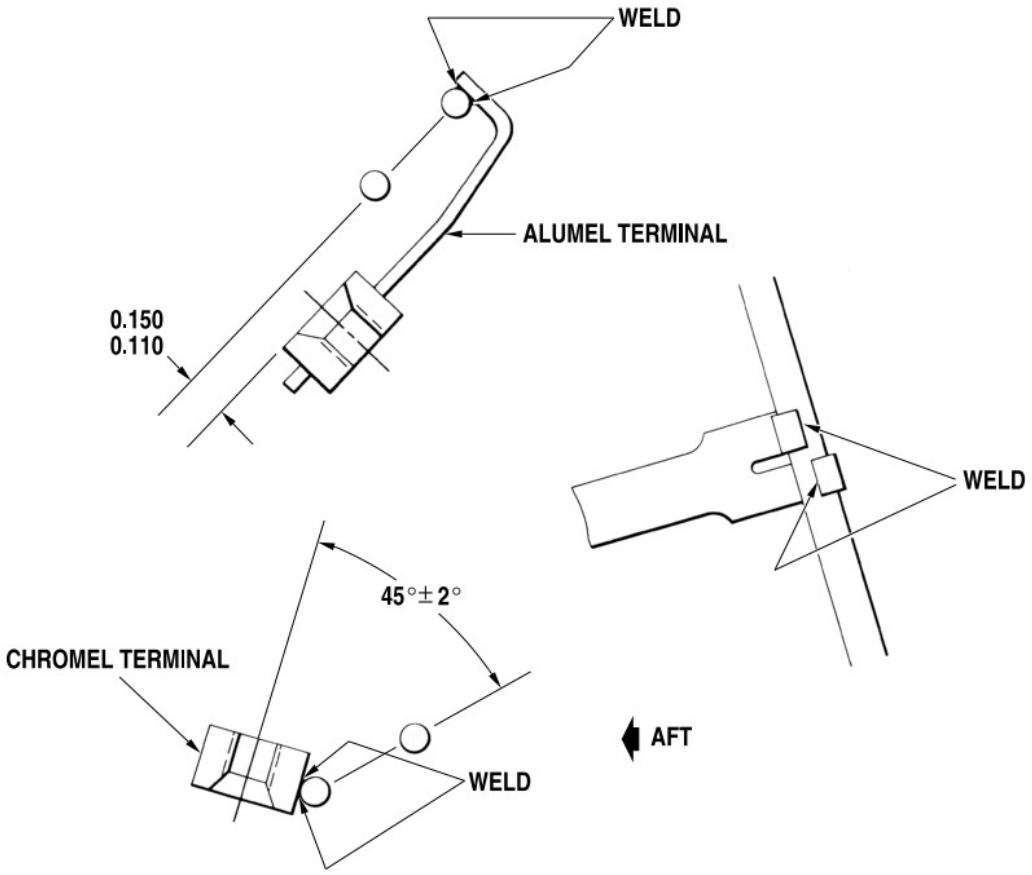
(4) Remove corrosion on the outer casing of the thermocouple with crocus cloth . Clean the trim harness lug contact surfaces with No. 400 grit abrasive cloth (PWC05-101).

9. Approved Repairs

A. Replacement of Harness Terminal Lugs (Ref. Fig. 204)

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Figure 204 Terminal Straps and Lug Replacement



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(1) Repair of the components of the interturbine temperature sensing system is limited to the replacement of missing insulation, the repair of loose or damaged terminal lugs and the repair of broken or deteriorated braid on the thermocouple wiring harness.

B. Replace Terminal Straps and Lugs on Bus-bar Assembly

(1) Suitably mark the position of the damaged terminal strap or lug on the bus-bar.

(2) Dress the bus-bar to remove the old weldment.

(3) Locate the replacement terminal strap on the bus-bar and, using the TIG method, fusion-weld the strap to the bus-bar. No filler rod is required.

(4) Locate the terminal lug on the bus-bar, making sure that the large chamfer on the threaded portion is facing outward (Ref. Fig. 204). Using the TIG method, fusion-weld the lug to the bus-bar. Use alumel (PWC05-164) or chromel (PWC05-353) welding filler rod as applicable.

NOTE: Regular "Type K" thermocouple wire (alumel or chromel), or suitable parts from rejected T5 bus-bar assemblies, may be used as an alternative filler material. Stress relief is not required.

C. Repair Missing Insulation on Terminal Block and Probes

WARNING: CARE MUST BE EXERCISED WHILE MIXING SYNTHETIC CERAMIC BINDER NO. 29 WITH FILLER NO. 29, OTHERWISE SPLASHES MIGHT CAUSE DISCOMFORT OF THE EYES. IN THE EVENT OF CONTACT WITH EYES, IMMEDIATELY WASH WITH WATER AND CONSULT A PHYSICIAN.

(1) Prepare the cement filler and binder (PWC08-009) to the desired consistency and in accordance with the manufacturer's instructions.

NOTE: The cement used for repair is supplied in two parts; a filler in powder form and a binder in liquid form.

(2) Clean the repair areas to make sure that all surfaces are free of grease and other contaminants. The presence of any contaminants will adversely affect the physical and electrical insulating properties of the cement mixture.

(3) Apply the cement mixture to replace the missing insulation.

(4) Allow the cement mixture to set. Initial setting should take place within 30 minutes. Do not handle or disturb the repair area during this time.

(5) Cure the cement mixture by allowing to air dry for 24 hours or, if a more rapid curing is desired, heat in the oven at 65°C (150°F) for three hours.

D. Repair of Loose Terminals in Terminal Block and Probe Housing

(1) Apply a few drops of the cement binder only (PWC08-009) at the junction of the loose terminal, and allow to set.

(2) Cure the assembly by allowing to air dry for 24 hours or, if a more rapid curing is desired, heat in the oven at 65 to 93°C (150 - 200°F) for 30 minutes.

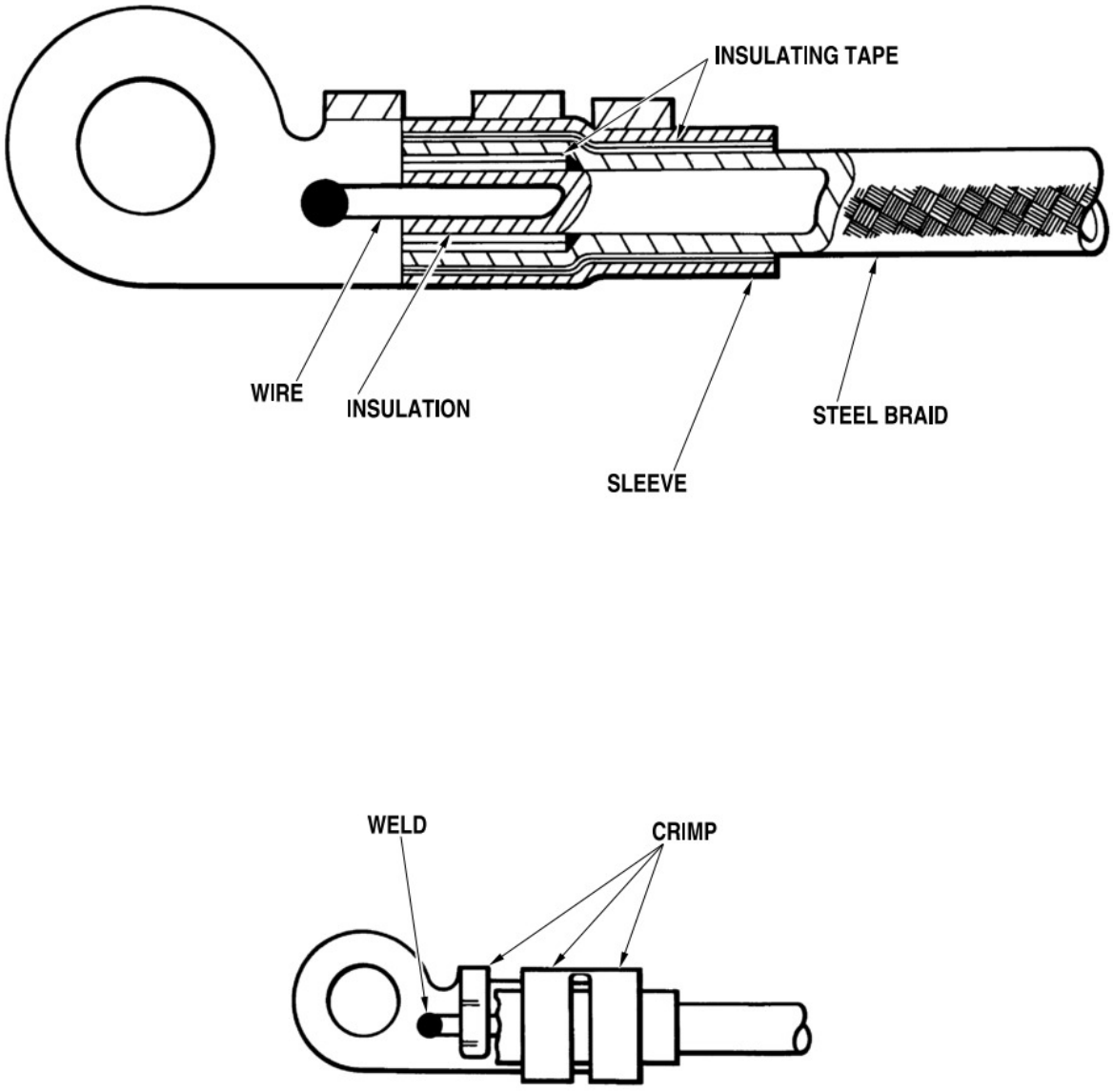
(3) Check the security of the terminal in the terminal block or probe housing; if the terminal is

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still loose, repeat the procedure.

E. Replacement of Damaged or Broken Harness Terminal Lugs (Ref. Figs. 205 and 206)

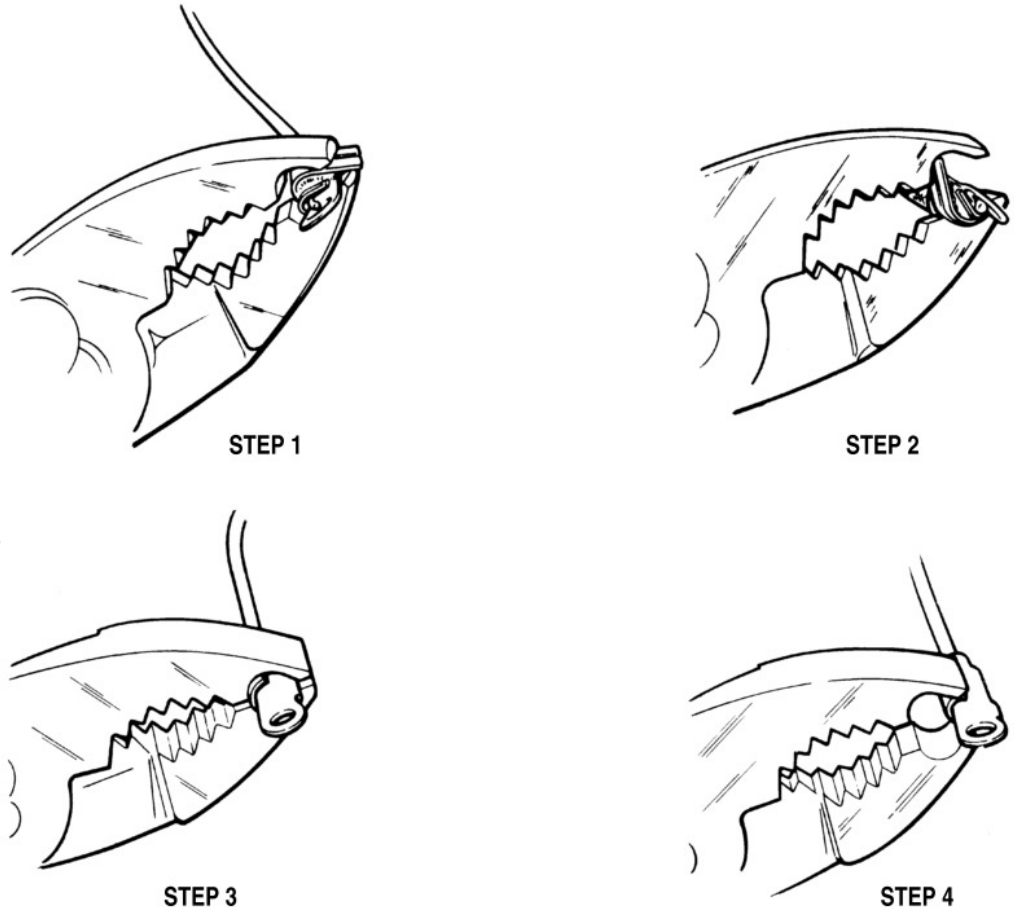
Figure 205 Wiring Harness Lug Replacement



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Figure 206 Harness Lug Replacement - Crimping



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NOTE: Harness leads must not be shortened by more than is absolutely necessary to carry out lug replacement in order to maintain harness resistance value within the acceptable limits.

(1) Open the ears of the terminal lug and cut the wire as close as possible to the weld. Remove the lug from the lead.

(2) Work the stainless steel braid back approximately 0.5 inch, taking care not to damage the underlying insulation.

(3) Wrap two complete turns of high-temperature insulating tape (PWC05-066) around the exposed harness insulation, and trim the frayed insulation back to 0.1875 inch from the end of the wire.

(4) Trim the stainless steel braid back to 0.3125 inch from the end of the wire, and wrap two complete turns of high-temperature insulating tape around the end of the braid.

(5) Slide the chafing sleeve (0.3125 inch long) over the insulating tape on the braid until the outer end of the sleeve is 0.0250 inch from the end of the wire. Use a twisting motion when fitting the sleeve to prevent the tape from unraveling.

(6) Secure a chromel or alumel terminal lug, as applicable, to the wire by crimping the small ear of the lug around the wire, leaving approximately 0.0625 inch of wire exposed.

(7) Use the TIG method (Ref. 70-00-00, REPAIR) to fusion-weld the wire to the terminal lug. Use alumel (PWC05-164) or chromel (PWC05-353) welding filler rod as applicable. Inspect and check the security of the weld by gently pulling the lug with pliers.

(8) Crimp the large ears of the terminal lug around the chafing sleeve (Ref. Fig. 205) on the braid using crimping pliers (PWC30425).

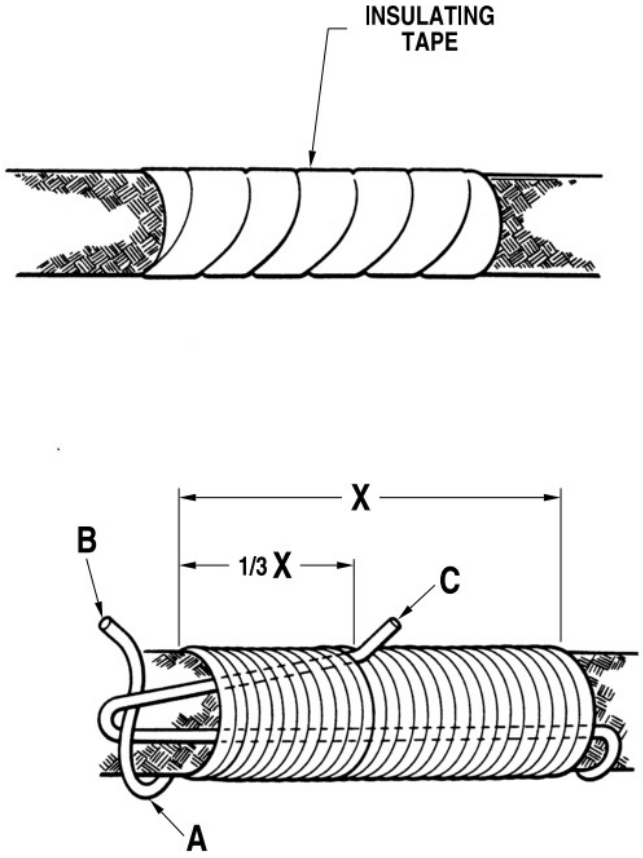
(9) Clean the contact surfaces of the terminal lug with abrasive cloth (PWC05-101), or a stainless steel wire brush.

(10) Do a loop resistance and continuity check (Ref. 77-20-00).

F. Repair of Harness Braiding (Ref. Fig. 207)

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Figure 207 Wiring Harness Braid Repair



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(1) Remove all loose strands of braiding around the damaged area.

(2) Wrap insulating tape (PWC05-066) around the harness, covering all the damaged area as shown.

(3) Wrap the repaired area with lockwire (PWC05-110) as shown. Hold the wrapping wire snugly along the length of the harness braid beginning from the left. Starting at the opposite end, wrap the wire tightly and uniformly over the harness braid and repaired area.

NOTE: Make sure the wire overlaps the tape and contacts the harness braid at each end of the repair.

(4) When approximately two-thirds of the wrapping length is completed, allow the short end of loop to remain exposed. Continue wrapping wire to complete the remaining distance, shown as "$^1/_3$ X".

(5) After the final turn of the wrapping, insert wire A through the loop as shown. Remove the surplus end of the wire B so the length is approximately $^1/_3$ of the repair length. Pull the loop end C until the wire at A is snug and the end of the wire B has just disappeared underneath the final turn of the wrappings. Cut the wire end C flush.

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